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(54) **METHOD FOR COMPENSATING ACTUATOR EFFECTS OF ACTUATORS**

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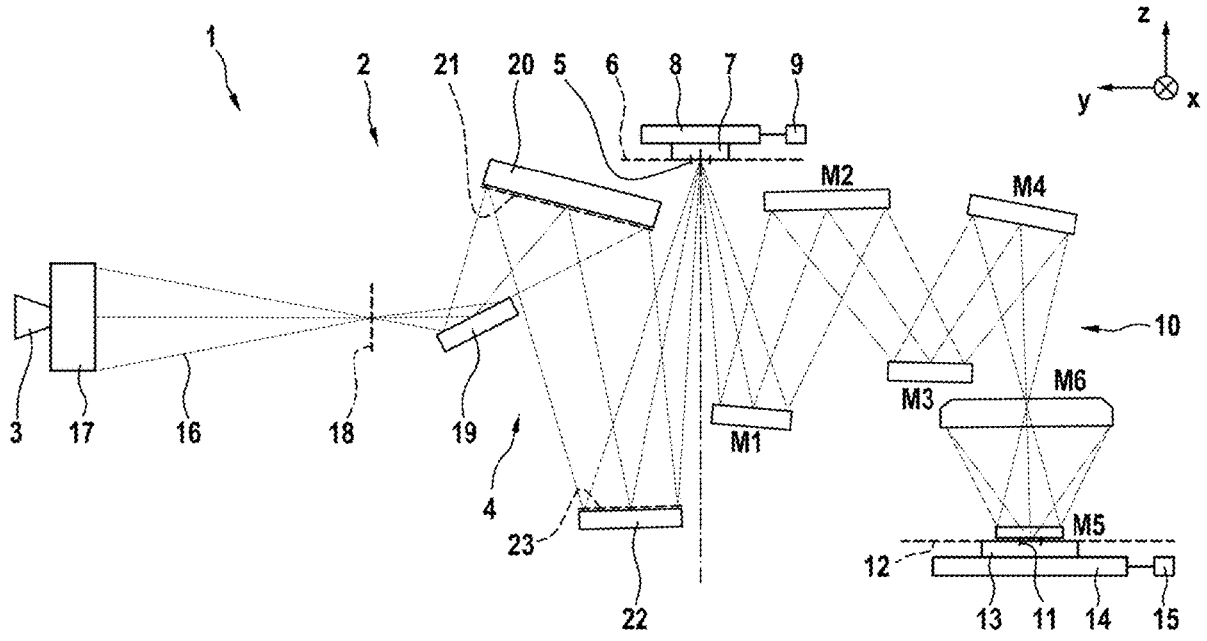
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(57) **ABSTRACT**

A method for driving an actuator for a component of a projection exposure apparatus for semiconductor lithography comprises: characterizing the actuator; parameterizing an actuator model; implementing the actuator model in a control structure; and driving the in actuator using the actuator model.



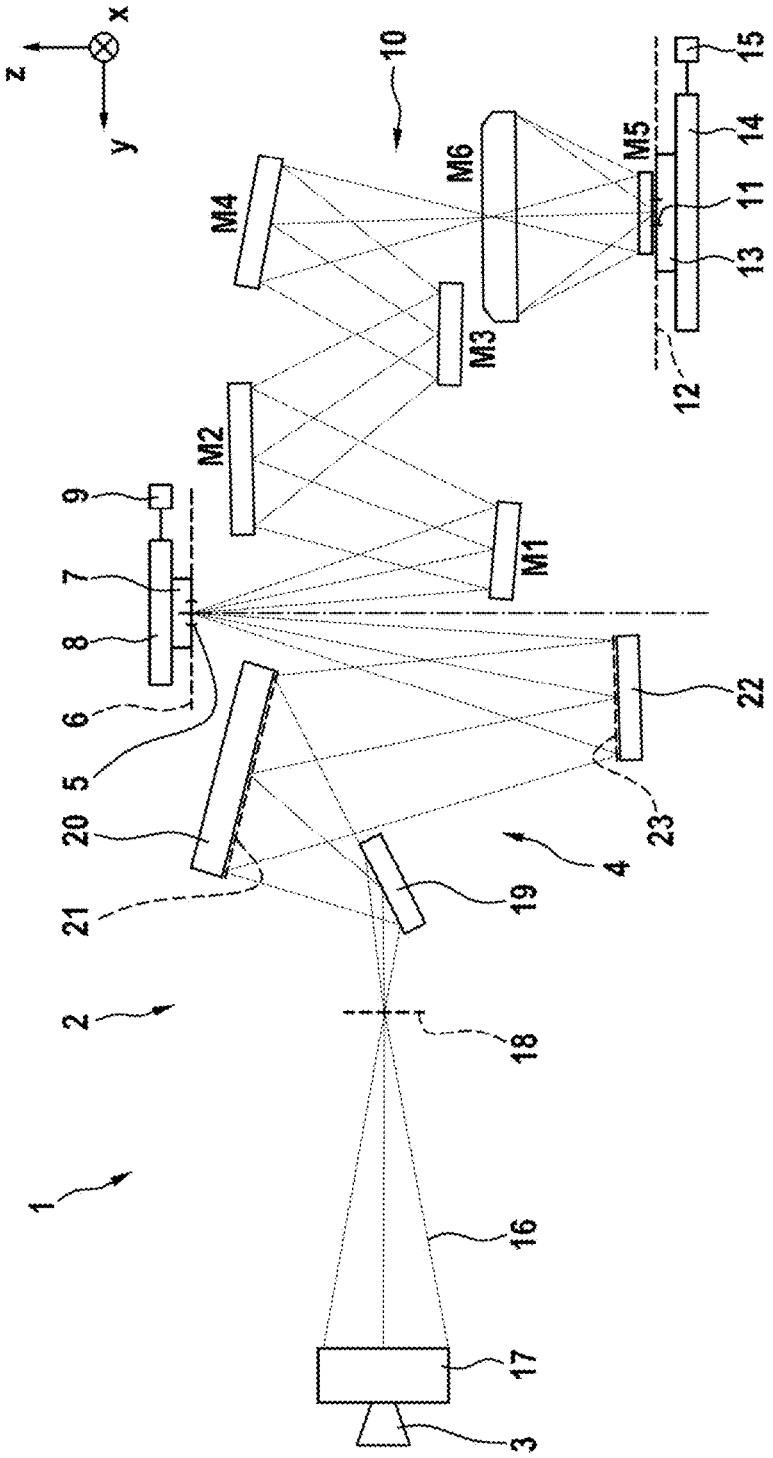

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Fig. 1

Fig. 2

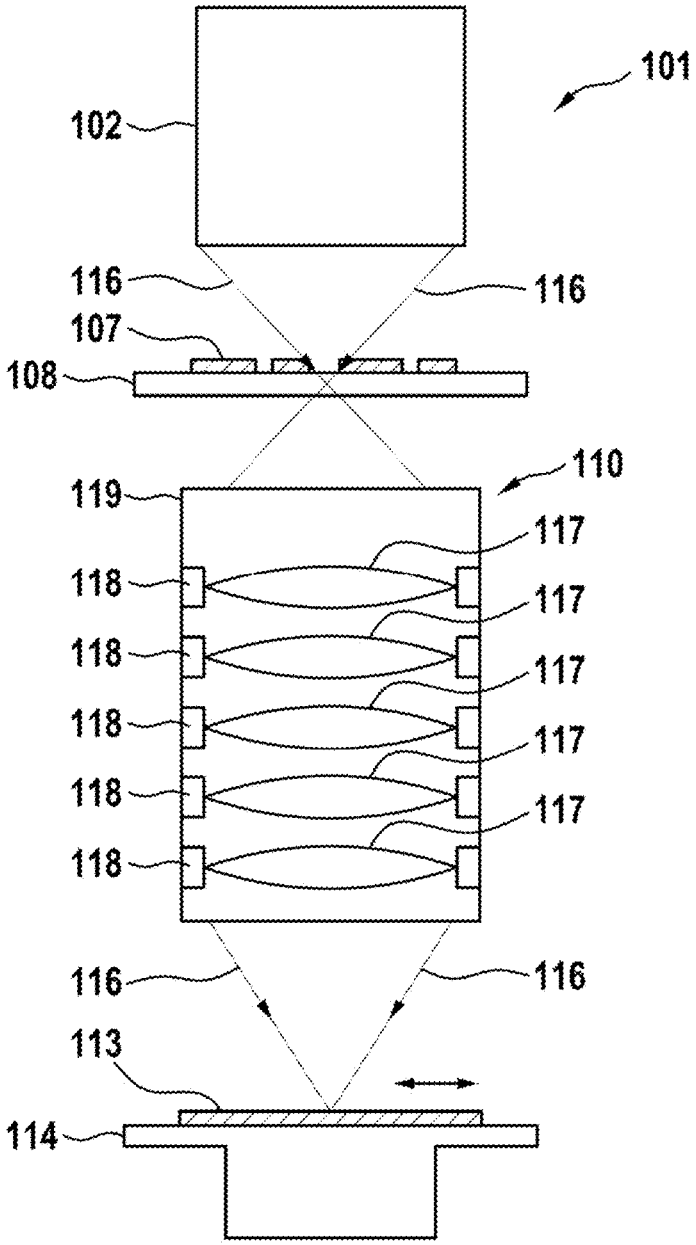

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Fig. 3

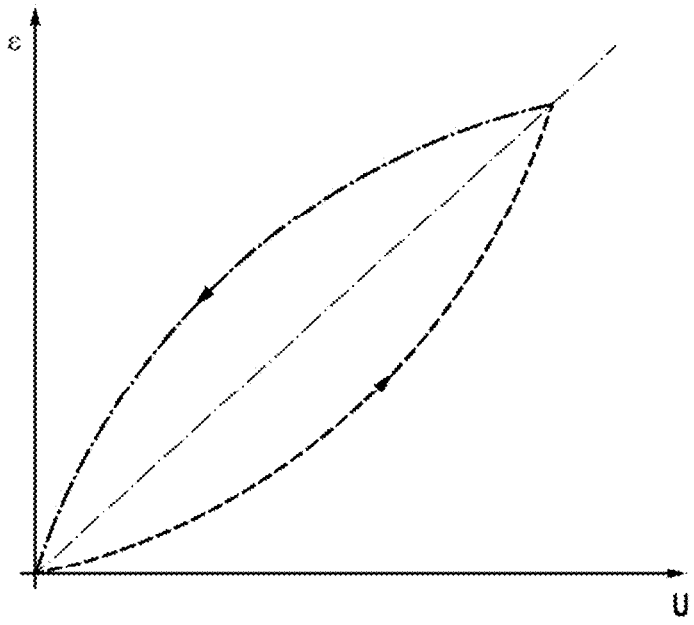


Fig. 4

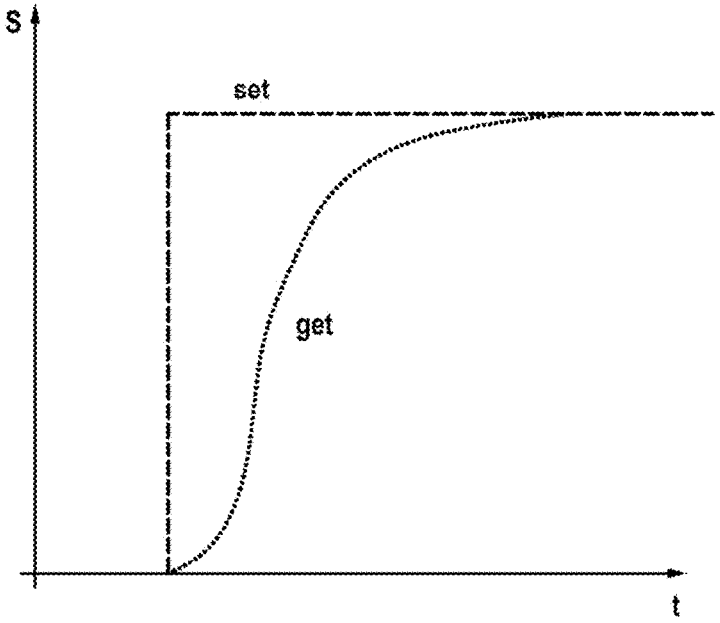


Fig. 5

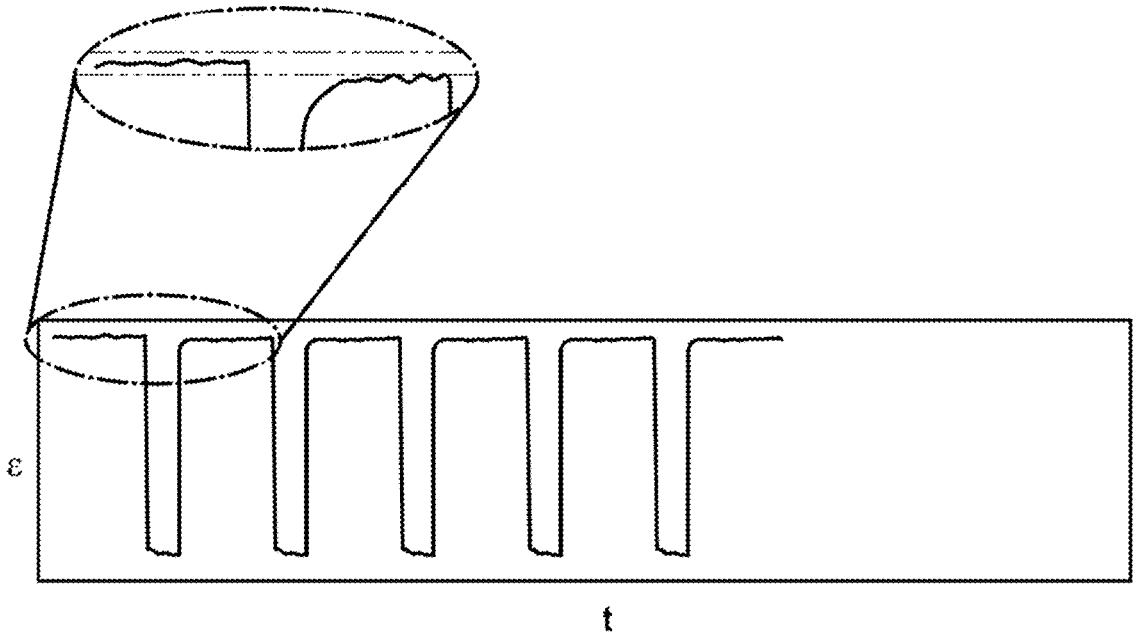


Fig. 6

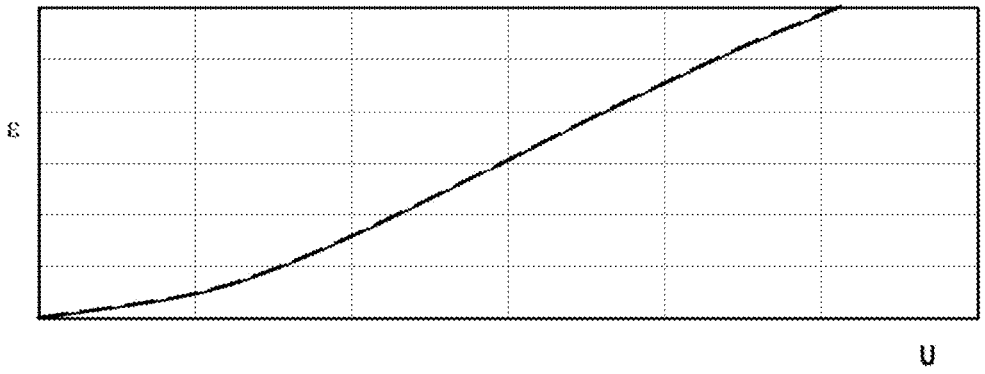


Fig. 7

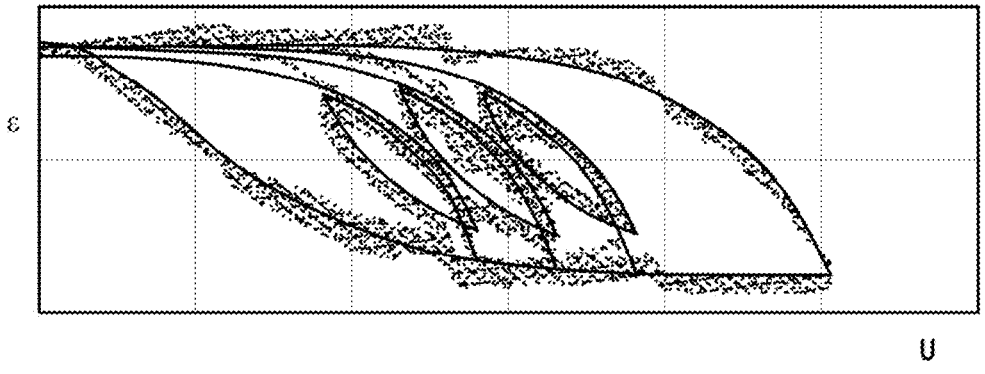


Fig. 8A

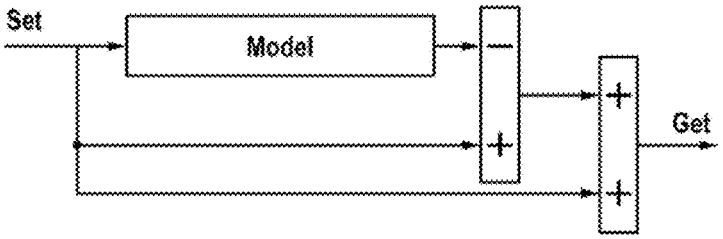


Fig. 8B



Fig. 8C

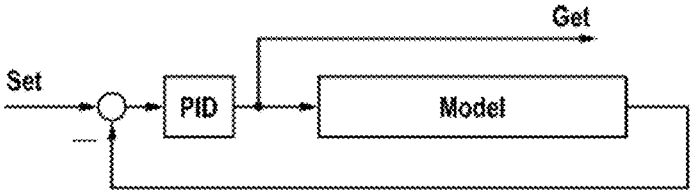


Fig. 8D

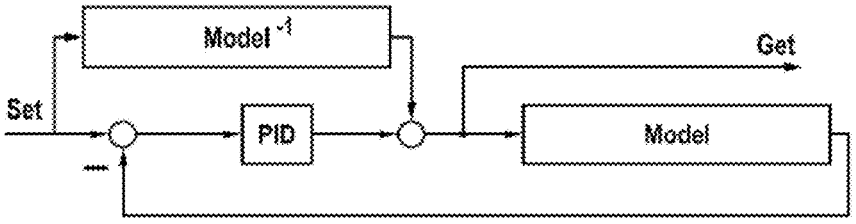


Fig. 9

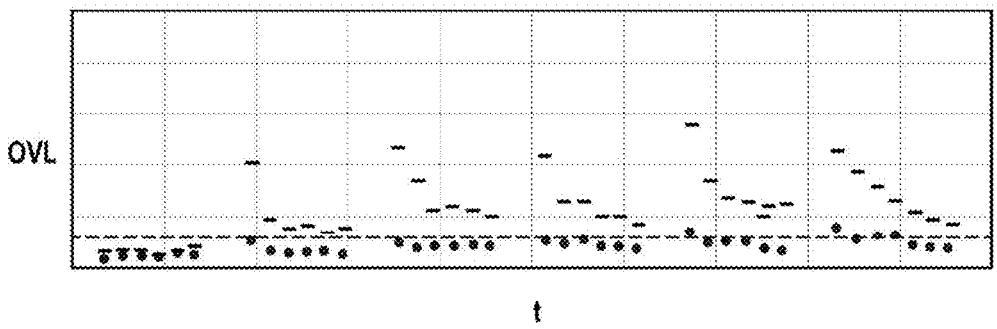
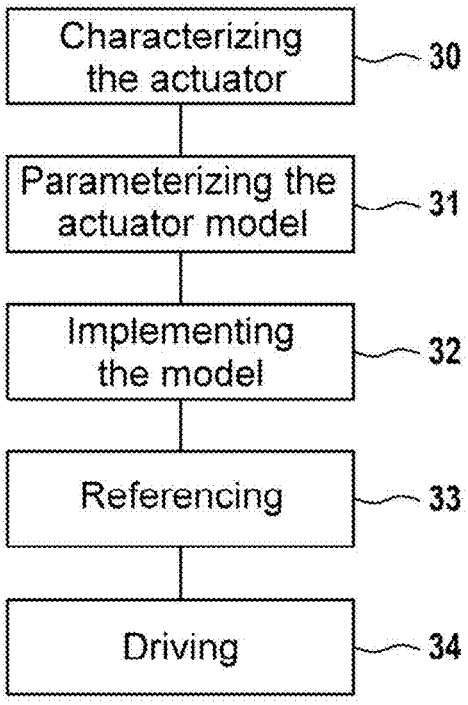

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Fig. 10


Characterizing the actuator
30
Parameterizing the actuator model
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Implementing the model
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Referencing
33
Driving
34

METHOD FOR COMPENSATING ACTUATOR EFFECTS OF ACTUATORS

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application is a continuation of, and claims benefit under 35 USC 120 to, international application No. PCT/EP2023/063340, filed May 17, 2023, which claims benefit under 35 USC 119 of German Application No. 10 2022 206 038.5, filed Jun. 15, 2022. The entire disclosure of each of these applications is incorporated by reference herein.

FIELD

[0002] The disclosure relates to a method for compensating actuator effects of actuators in projection exposure apparatuses for semiconductor lithography.

BACKGROUND

[0003] Adaptive optical elements are becoming ever more relevant as the demands on lithography systems, especially in the DUV or EUV range, increase. For example, such elements may be in the form of deformable mirrors which can be driven within extremely short periods of time by actuators, for example, in order to compensate for a wavefront aberration as a result of a deformation of an optically effective surface of the mirror. The optically effective surface is the surface of the mirror on which the light used to image structures of a mask onto a wafer is incident during the normal operation of the apparatus.

[0004] Electrostrictive actuators, or else piezoactuators, which are classed among the ferro electric solid-state actuators, are frequently used as actuators. However, such actuators frequently exhibit unwanted effects, for example hysteresis and creep effects. This behaviour of the actuators can be especially damaging if what is known as a feed-forward method is used for driving the actuator system. The aforementioned method is distinguished in that it is merely an actuating signal generated in a control unit that is output to the actuator for the purpose of setting a desired state, for example the deflection of an actuator. The response of the system, which is to say the path actually travelled by the actuator, initially remains unconsidered for this type of control. Thus, very stringent demands are usually placed upon the model forming the basis for the calculation and output of the control signal for a desired deflection of the actuator.

SUMMARY

[0005] The present disclosure seeks to provide a method by which it is possible to attain improved accuracy for driving actuators in projection exposure apparatuses for semiconductor lithography.

[0006] The disclosure provides a method for driving an actuator for an optical component of a projection exposure apparatus for semiconductor lithography, the method comprising:

[0007] characterizing the actuator;

[0008] parameterizing an actuator model;

[0009] implementing the actuator model in a control structure; and

[0010] driving the actuator using the actuator model.

[0011] According to the disclosure, as a result of the properties of the respective actuator finding consideration by way of the application of the actuator model when driving the actuator, it is possible to minimize an influence of certain undesirable effects on the precision of the actuator control. For example, this can bring about an improvement in the imaging quality, for example in the overlay performance of a projection exposure apparatus.

[0012] A reference step is performed at certain times in an embodiment of the disclosure. In this context, a reference step should be understood to mean a method step in which a defined state of the considered system is established. In this case, the considered system may comprise, for example, the model itself and its parameters, but also the real world, for example an actuator.

[0013] Thus, for example, the reference step can make it possible to take into consideration the circumstance that even a model merely present in software is subject to changes over a long period of time. For example, parameters set at the start of the method according to the disclosure may change, purely on account of the nature of the computer hardware, over a period of several days, weeks or years. Accordingly, it may be desirable to reset these parameters intermittently.

[0014] Moreover, it can be desirable to also intermittently put the driven actuator into a defined deflection state. As already mentioned above, the starting point and the direction in which a control voltage is applied definitely plays a role in the real actuator deflection. On the basis thereof, the real deflection of the actuator may travel along one of the two branches of a hysteresis curve. If a defined state is now set by way of an appropriately defined setting of the actuator deflection and the suitable choice of the subsequent driving direction, it is possible to ensure being on the correct branch of the hysteresis curve.

[0015] The period of time between the exposure of two wafers represents a choice for the time of the reference step. The few milliseconds of time available here are sufficient to perform the desired referencing. In this context, the referencing itself need not always comprise a resetting of the model and the control of the actuator; naturally, it is also conceivable to carry out only one of the two measures.

[0016] For example, one or more of the following actuator parameters lend themselves to the characterization of the actuator for the purpose of preparing the method according to the disclosure: change in length, frequency response, hysteresis, drift.

[0017] For example, the actuator can be characterized within a test environment. In this case, it is also conceivable to use not the actuator itself but a comparable sample for the characterization.

[0018] In an alternative, the actuator can be characterized in a projection exposure apparatus. One or more wafers can be exposed in this case, for example within a test run. The exposed wafers are subsequently measured. Then, the actuator parameters can likewise be determined from the ascertained image aberrations if a suitable model for the relationship between image aberrations and actuator properties is used.

[0019] In a variant of the disclosure, at least one separate model for at least one of the actuator parameters, which is subsequently superposed on at least one further model, is generated during the parameterization of the actuator model.

[0020] In this context, models for the drift can include:

[0021] Padde approximation;

$$S = a * \tanh(b * U)^2 + c;$$

$$P = \tanh(b * U + c * P + d * P^3);$$

[0022] $S=a*P^2$; and

[0023] polynomials;

[0024] where S represents the actuator deflection, P represents the surface charge density, U represents the voltage, and a, b, c, d are fit parameters.

[0025] The following models, inter alia, are suitable for describing the hysteresis:

[0026] Bouc Wen;

[0027] Prandtl-Ishlinskii;and

[0028] Preisach.

[0029] Models for modelling the dynamics include inter alia:

[0030] superposed Pt1 functions;

[0031] superposed log functions; and

[0032] fractional differential equations.

[0033] A model developed as described above can be subsequently implemented in a controller which can be used to drive the actuator system of a component of a projection exposure apparatus.

[0034] For example the following options may be chosen for control purposes:

[0035] inverse model;

[0036] model-based control;

[0037] observer-based control;

[0038] subtracting of the actuator model;

[0039] machine learning control; and

[0040] neural network control.

[0041] For example, the actuator can be an electrostrictive actuator and a piezoelectric actuator or a magnetostrictive actuator.

[0042] For example, the actuator may be configured to position and also/or else deform the component.

[0043] The component can be an optical element, such as a mirror, optionally a deformable mirror.

BRIEF DESCRIPTION OF THE DRAWINGS

[0044] Exemplary embodiments and variants of the disclosure will be explained in more detail hereinafter with reference to the drawings, in which:

[0045] FIG. **1** schematically shows a meridional section of a projection exposure apparatus for EUV projection lithography;

[0046] FIG. **2** schematically shows a meridional section of a projection exposure apparatus for DUV projection lithography;

[0047] FIG. **3** shows a typical actuator hysteresis curve;

[0048] FIG. **4** shows a typical drift behaviour of an actuator;

[0049] FIG. **5** shows a drift in the direction of the target deflection of an exemplary actuator;

[0050] FIG. **6** shows the deflection of an actuator counter to a voltage applied thereto;

[0051] FIG. **7** shows a consideration of the differences between the values obtained from the model and measurement;

[0052] FIGS. **8**A-**8**D show different variants for implementing the model;

[0053] FIG. **9** shows the effect of the method according to the disclosure; and

[0054] FIG. **10** shows a flowchart for the method according to the disclosure.

DETAILED DESCRIPTION

[0055] Certain parts of a microlithographic projection exposure apparatus **1** are described in exemplary fashion below initially with reference to FIG. **1**. The description of the fundamental structure of the projection exposure apparatus **1** and the integral parts thereof is understood here to be non-limiting.

[0056] An embodiment of an illumination system **2** of the projection exposure apparatus **1** has, in addition to a radiation source **3**, an illumination optical unit **4** for illuminating an object field **5** in an object plane **6**. In an alternative embodiment, the light source **3** may also be provided as a module separate from the rest of the illumination system. In this case, the illumination system does not comprise the light source **3**.

[0057] A reticle **7** arranged in the object field **5** is illuminated. The reticle **7** is held by a reticle holder **8**. The reticle holder **8** is displaceable by way of a reticle displacement drive **9** in particular in a scanning direction.

[0058] A Cartesian xyz-coordinate system is shown in FIG. **1** for explanation purposes. The x-direction runs perpendicular to the plane of the drawing into the latter. The y-direction runs horizontally, and the z-direction runs vertically. The scanning direction runs in the y-direction in FIG. **1**. The z-direction runs perpendicular to the object plane **6**.

[0059] The projection exposure apparatus **1** comprises a projection optical unit **10**. The projection optical unit **10** serves for imaging the object field **5** into an image field **11** in an image plane **12**. The image plane **12** extends parallel to the object plane **6**. Alternatively, an angle that differs from 0° between the object plane **6** and the image plane **12** is also possible.

[0060] A structure on the reticle **7** is imaged onto a light-sensitive layer of a wafer **13** arranged in the region of the image field **11** in the image plane **12**. The wafer **13** is held by a wafer holder **14**. The wafer holder **14** is displaceable by way of a wafer displacement drive **15**, for example in the y-direction. The displacement, firstly, of the reticle **7** by way of the reticle displacement drive **9** and, secondly, of the wafer **13** by way of the wafer displacement drive **15** may be implemented so as to be mutually synchronized.

[0061] The radiation source **3** is an EUV radiation source. The radiation source **3** emits EUV radiation **16**, which is also referred to below as used radiation, illumination radiation or illumination light. For example, the used radiation has a wavelength in the range of between 5 nm and 30 nm. The radiation source **3** can be a plasma source, for example an LPP (laser produced plasma) source or a GDPP (gas discharge produced plasma) source. It may also be a synchrotron-based radiation source. The radiation source **3** may be a free electron laser (FEL).

[0062] The illumination radiation **16** emerging from the radiation source **3** is focused by a collector **17**. The collector **17** may be a collector with one or more ellipsoidal and/or hyperboloidal reflection surfaces. The illumination radiation **16** can be incident on the at least one reflection surface of the collector **17** with grazing incidence (GI), which is to say at

angles of incidence of greater than 45° relative to the direction of the normal to the mirror surface, or with normal incidence (NI), which is to say at angles of incidence of less than 45°. The collector **17** can be structured and/or coated firstly for optimizing its reflectivity for the used radiation and secondly for suppressing extraneous light.

[0063] Downstream of the collector **17**, the illumination radiation **16** propagates through an intermediate focus in an intermediate focal plane **18**. The intermediate focal plane **18** can represent a separation between a radiation source module, having the radiation source **3** and the collector **17**, and the illumination optical unit **4**.

[0064] The illumination optical unit **4** comprises a deflection mirror **19** and, arranged downstream thereof in the beam path, a first facet mirror **20**. The deflection mirror **19** may be a plane deflection mirror or, alternatively, a mirror with a beam-influencing effect going beyond a pure deflection effect. Alternatively or in addition, the deflection mirror **19** may be in the form of a spectral filter that separates a used light wavelength of the illumination radiation **16** from extraneous light of a wavelength deviating therefrom. If the first facet mirror **20** is arranged in a plane of the illumination optical unit **4** that is optically conjugate to the object plane **6** as a field plane, it is also referred to as a field facet mirror. The first facet mirror **20** comprises a multiplicity of individual first facets **21**, which are also referred to below as field facets. FIG. **1** depicts only some of the facets **21** by way of example.

[0065] The first facets **21** can be embodied in the form of macroscopic facets, for example in the form of rectangular facets or in the form of facets with an arcuate peripheral contour or a peripheral contour of part of a circle. The first facets **21** may be embodied as plane facets or alternatively as facets with convex or concave curvature.

[0066] As known for example from DE 10 2008 009 600 A1, the first facets **21** themselves can also be composed in each case of a multiplicity of individual mirrors, such as a multiplicity of micromirrors. For example, the first facet mirror **20** can be a microelectromechanical system (MEMS system). For details, reference is made to DE 10 2008 009 600 A1.

[0067] The illumination radiation **16** travels horizontally, which is to say in the y-direction, between the collector **17** and the deflection mirror **19**.

[0068] In the beam path of the illumination optical unit **4**, a second facet mirror **22** is arranged downstream of the first facet mirror **20**. If the second facet mirror **22** is arranged in a pupil plane of the illumination optical unit **4**, it is also referred to as a pupil facet mirror. The second facet mirror **22** can also be arranged at a distance from a pupil plane of the illumination optical unit **4**. In this case, the combination of the first facet mirror **20** and the second facet mirror **22** is also referred to as a specular reflector. Specular reflectors are known from US 2006/0132747 A1, EP 1 614 008 B1, and U.S. Pat. No. 6,573,978.

[0069] The second facet mirror **22** comprises a plurality of second facets **23**. In the case of a pupil facet mirror, the second facets **23** are also referred to as pupil facets.

[0070] The second facets **23** may likewise be macroscopic facets, which may for example have a round, rectangular or hexagonal boundary, or may alternatively be facets composed of micromirrors. In this regard, reference is likewise made to DE 10 2008 009 600 A1.

[0071] The second facets **23** may have plane reflection surfaces or alternatively convexly or concavely curved reflection surfaces.

[0072] The illumination optical unit **4** consequently forms a doubly faceted system. This fundamental principle is also referred to as a fly's eye condenser (fly's eye integrator).

[0073] It can be desirable to arrange the second facet mirror **22** not exactly in a plane that is optically conjugate to a pupil plane of the projection optical unit **10**. For example, the pupil facet mirror **22** can be arranged so as to be tilted relative to a pupil plane of the projection optical unit **10**, as is described, for example, in DE 10 2017 220 586 A1.

[0074] The individual first facets **21** are imaged into the object field **5** with the aid of the second facet mirror **22**. The second facet mirror **22** is the last beam-shaping mirror or indeed the last mirror for the illumination radiation **16** in the beam path upstream of the object field **5**.

[0075] In a further embodiment (not shown) of the illumination optical unit **4**, a transfer optical unit contributing in particular to the imaging of the first facets **21** into the object field **5** may be arranged in the beam path between the second facet mirror **22** and the object field **5**. The transfer optical unit may have exactly one mirror, or alternatively have two or more mirrors, which are arranged one behind the other in the beam path of the illumination optical unit **4**. The transfer optical unit may for example comprise one or two normal-incidence mirrors (NI mirrors) and/or one or two grazing-incidence mirrors (GI mirrors).

[0076] In the embodiment shown in FIG. **1**, the illumination optical unit **4** has exactly three mirrors downstream of the collector **17**, specifically the deflection mirror **19**, the field facet mirror **20**, and the pupil facet mirror **22**.

[0077] The deflection mirror **19** can also be dispensed with in a further embodiment of the illumination optical unit **4**, and so the illumination optical unit **4** can then have exactly two mirrors downstream of the collector **17**, specifically the first facet mirror **20** and the second facet mirror **22**.

[0078] The imaging of the first facets **21** into the object plane **6** via the second facets **23** or using the second facets **23** and a transfer optical unit is, in general, only approximate imaging.

[0079] The projection optical unit **10** comprises a plurality of mirrors Mi, which are consecutively numbered in accordance with their arrangement in the beam path of the projection exposure apparatus **1**.

[0080] In the example illustrated in FIG. **1**, the projection optical unit **10** comprises six mirrors M**1** to M**6**. Alternatives with four, eight, ten, twelve or any other number of mirrors Mi are likewise possible. The penultimate mirror M**5** and the last mirror M**6** each have a passage opening for the illumination radiation **16**. The projection optical unit **10** is a doubly obscured optical unit. The projection optical unit **10** has an image-side numerical aperture which is greater than 0.5 and which can also be greater than 0.6 and, for example, can be 0.7 or 0.75.

[0081] Reflection surfaces of the mirrors Mi can be embodied as freeform surfaces without an axis of rotational symmetry. Alternatively, the reflection surfaces of the mirrors Mi can be designed as aspherical surfaces with exactly one axis of rotational symmetry of the reflection surface shape. Just like the mirrors of the illumination optical unit **4**, the mirrors Mi can have highly reflective coatings for the illumination radiation **16**.

[0082] These coatings can be designed as multilayer coatings, for example with alternating layers of molybdenum and silicon.

[0083] The projection optical unit **10** has a large object-image offset in the y-direction between a y-coordinate of a centre of the object field **5** and a y-coordinate of the centre of the image field **11**. This object-image offset in the y-direction can be of approximately the same magnitude as a z-distance between the object plane **6** and the image plane **12**.

[0084] For example, the projection optical unit **10** can have an anamorphic embodiment. For example, it can have different imaging scales βx, βy in the x- and y-directions. The two imaging scales Bx, By of the projection optical unit **10** can be (βx, βy)=(+/−0.25, +/−0.125). A positive imaging scale β means imaging without image inversion. A negative sign for the imaging scale β means imaging with image inversion.

[0085] The projection optical unit **10** consequently leads to a reduction in size with a ratio of 4:1 in the x-direction, which is to say in a direction perpendicular to the scanning direction. The projection optical unit **10** leads to a reduction in size of 8:1 in the y-direction, which is to say in the scanning direction.

[0086] Other imaging scales are likewise possible. Imaging scales with the same sign and the same absolute value in the x-direction and y-direction are also possible, for example with absolute values of 0.125 or of 0.25.

[0087] The number of intermediate image planes in the x-direction and in the y-direction in the beam path between the object field **5** and the image field **11** can be the same or can differ depending on the embodiment of the projection optical unit **10**. Examples of projection optical units with different numbers of such intermediate images in the x and y-directions are known from US 2018/0074303 A1.

[0088] In each case one of the pupil facets **23** is assigned to exactly one of the field facets **21** for forming in each case an illumination channel for illuminating the object field **5**. For example, this can yield illumination according to the Köhler principle. The far field is decomposed into a multiplicity of object fields **5** with the aid of the field facets **21**. The field facets **21** generate a plurality of images of the intermediate focus on the pupil facets **23** respectively assigned thereto.

[0089] By way of an assigned pupil facet **23**, the field facets **21** are imaged in each case onto the reticle **7** in a manner overlaid on one another for the purposes of illuminating the object field **5**. The illumination of the object field **5** is, for example, as homogeneous as possible. It can have a uniformity error of less than 2%. The field uniformity can be achieved by overlaying different illumination channels.

[0090] The illumination of the entrance pupil of the projection optical unit **10** can be defined geometrically by way of an arrangement of the pupil facets. The intensity distribution in the entrance pupil of the projection optical unit **10** can be set by selecting the illumination channels, in particular the subset of the pupil facets which guide light. This intensity distribution is also referred to as illumination setting.

[0091] A likewise preferred pupil uniformity in the region of sections of an illumination pupil of the illumination optical unit **4** that are illuminated in a defined manner can be achieved by a redistribution of the illumination channels.

[0092] Further aspects and details of the illumination of the object field **5** and for example of the entrance pupil of the projection optical unit **10** are described below.

[0093] The projection optical unit **10** may have a homocentric entrance pupil. The latter can be accessible. It can also be inaccessible.

[0094] The entrance pupil of the projection optical unit **10** cannot, as a rule, be exactly illuminated using the pupil facet mirror **22**. The aperture rays often do not intersect at a single point when imaging the projection optical unit **10** which telecentrically images the centre of the pupil facet mirror **22** onto the wafer **13**. However, it is possible to find an area in which the spacing of the aperture rays that is determined in pairs becomes minimal. This area represents the entrance pupil or an area in real space that is conjugate thereto. For example, this area has a finite curvature.

[0095] It may be the case that the projection optical unit **10** has different poses of the entrance pupil for the tangential beam path and for the sagittal beam path. In this case, an imaging element, in particular an optical component part of the transfer optical unit, should be provided between the second facet mirror **22** and the reticle **7**. With the aid of this optical element, the different poses of the tangential entrance pupil and **20** the sagittal entrance pupil can be taken into account.

[0096] In the arrangement of the components of the illumination optical unit **4** illustrated in FIG. **1**, the pupil facet mirror **22** is arranged in an area conjugate to the entrance pupil of the projection optical unit **10**. The field facet mirror **20** is arranged so as to be tilted with respect to the object plane **6**. The first facet mirror **20** is arranged so as to **25** be tilted with respect to an arrangement plane defined by the deflection mirror **19**.

[0097] The first facet mirror **20** is arranged so as to be tilted with respect to an arrangement plane defined by the second facet mirror **22**.

[0098] FIG. **2** schematically shows a meridional section of a further projection exposure apparatus **101** for DUV projection lithography, in which the disclosure can likewise be used.

[0099] The structure of the projection exposure apparatus **101** and the principle of the imaging are comparable with the construction and procedure described in FIG. **1**. Identical component parts are denoted by a reference sign increased by **100** relative to FIG. **1**, which is to say the reference signs in FIG. **2** begin with **101**.

[0100] In contrast to an EUV projection exposure apparatus **1** as described in FIG. **1**, refractive, diffractive and/or reflective optical elements **117**, such as for example lens elements, mirrors, prisms, terminating plates, and the like, can be used for imaging or for illumination in the DUV projection exposure apparatus **101** on account of the greater wavelength of the DUV radiation **116**, employed as used light, in the range from 100 nm to 300 nm, such as of 193 nm. The projection exposure apparatus **101** in this case comprises an illumination system **102**, a reticle holder **108** for receiving and exactly positioning a reticle **107** provided with a structure, by which the later structures on a wafer **113** are determined, a wafer holder **114** for holding, moving, and exactly positioning the wafer **113**, and a projection lens **110**, with a plurality of optical elements **117**, which are held by way of mounts **118** in a lens housing **119** of the projection lens **110**.

[0101] The illumination system **102** provides DUV radiation **116** for imaging the reticle **107** on the wafer **113**. A laser, a plasma source or the like can be used as the source of this radiation **116**. The radiation **116** is shaped in the illumination system **102** via optical elements such that the DUV radiation **116** has the desired properties with regard to diameter, polarization, shape of the wavefront and the **25** like when it is incident on the reticle **107**.

[0102] Apart from the additional use of refractive optical elements **117**, such as lens elements, prisms, terminating plates, the construction of the downstream projection optical unit **110** with the lens housing **119** does not differ in principle from the construction described in FIG. **1** and is therefore not described in further detail.

[0103] The apparatuses shown in FIGS. **1** and **2** each contain a multiplicity of components which can be positioned or else deformed via actuators. Thus, the properties of the actuators used to this end have an immediate effect on the performance of the apparatuses shown.

[0104] FIG. **3** schematically shows the behaviour of the mechanical strain of an actuator when a voltage is applied, until a maximum voltage is reached, and a subsequent retraction to a minimum voltage, 0 V in the example shown. This behaviour is known per se. The occurrence of a hysteresis is easily identifiable from the figure. In other words, the mechanical strain of the actuator for corresponding voltage values in the case of a fall in voltage does not correspond to the mechanical strain of the actuator in the case of an increase in voltage. In general, it is hard to explain the underlying effects from a physical point of view and these effects can only be modelled macroscopically.

[0105] Furthermore, as illustrated in FIG. **4**, a typical actuator also has a certain drift behaviour in addition to the above-described effect of hysteresis. In other words, the desired deflection of an actuator is not set instantaneously, but with a certain time profile, even if a rectangular voltage signal is applied.

[0106] The aforementioned effects can be relevant if an actuator is to accomplish a significant deflection within a comparatively short period of time. This situation is elucidated on the basis of FIG. **5**. The drift in the direction of the target deflection of an exemplary actuator during the first exposures of adjacent regions following a deflection of the actuator over a relatively large travel is easily identifiable in the figure. The assumption can be made that only an unsatisfactory imaging result can be obtained for these first regions.

[0107] The mode of operation of the model should be explained in exemplary fashion on the basis of FIGS. **6** and **7** below.

[0108] In this case, the deflection of an actuator is plotted qualitatively against a voltage applied thereto in the illustration shown in FIG. **6**. In this case, the real deflection is represented by a dashed curve while the curve generated on the basis of the model is represented by a dotted curve. The initially small deviation between the two curves is easily identifiable in the figure.

[0109] The quality and mode of operation of the model is only rendered identifiable if a consideration of the differences, as illustrated in FIG. **7**, is carried out. The solid curve in FIG. **7** illustrates the result of subtracting from a curve generated on the basis of the complete model, which is to say under consideration of hysteresis and drift in particular, the curve which arises when a model that does not consider the aforementioned effects is used. This results in an illustration which clearly identifies the influence of the effects. As already mentioned, the solid curve is based only on a 10 model-based calculation.

[0110] The second graph illustrated in FIG. 7 arises from subtracting actually measured values for the deflection of the actuator in question from the ideal curve, which is to say the curve that would arise from the model if the effects are omitted. The high degree of correspondence between the two representations, which allows conclusions **15** to be drawn about the quality of the model used, is clearly identifiable in the figure.

[0111] The illustrations shown in FIGS. **8**A-**8**D show different variants for implementing the model.

[0112] Here, FIG. **8**A initially visualizes a subtraction method. This method is substantially based on the fact that initially a certain target value of a deflection, for example 40 pm, is assumed. Subsequently, the model is used to determine the simulated deflection that would set-in under the assumption of the aforementioned target value. For example, if the model now supplies a value of 38 pm, then the difference of the two values, which is to say 2 pm, is applied to the target value which then forms the basis for the real control, in order to obtain the desired deflection.

[0113] FIG. **8**B illustrates an inversion method. In this case, output and input are interchanged in the case of an invertible model and integrated into the controller.

[0114] FIG. **8**C illustrates a model-based closed-loop control method. In this case, the planned actuating signal is initially used as starting value for the model, which subsequently supplies a certain travel. The travel simulated by the model is then supplied to a loop controller which performs a comparison between the desired model and the model obtained thereby, and subsequently adjusts the actuating signal, once again using the model, until the desired travel arises. As soon as this state is obtained, the actuating signal obtained in this manner is used to drive the actuator.

[0115] FIG. **8**D shows a combination of the variants illustrated on the basis of FIGS. **8**B and **8**C. Naturally, combinations deviating therefrom are also possible.

[0116] The effect of the method according to the disclosure is illustrated one more time on the basis of FIG. **9**. Here, the overlay error of generated structures is plotted over time, in each case following the deflection of the actuator. Here, the individual exposure procedures are visualized using dots for the corrected case and dashes for the uncorrected case. For the uncorrected case, the significant error for the respective first exposure procedures of a group is clearly evident from the figure. As already explained above, this is due to the fact that, following a displacement over a significant travel, the actuator has not yet reached its sought end position at the beginning of the group. In contrast thereto, practically no deviation can be identified in the corrected case (illustrated using dots).

[0117] FIG. **10** shows a method procedure according to the disclosure schematically in a flowchart.

[0118] The following steps are illustrated:

- **[0119]** characterizing **30** the actuator;
- **[0120]** parameterizing **31** an actuator model;
- **[0121]** implementing **32** the actuator model in a control structure;
- **[0122]** driving **34** the actuator using the actuator model; and
- **[0123]** the described referencing **33**.

[0124] It is self-evident that the schematic illustration shown in FIG. **10** is purely exemplary. For example, the referencing **33** is an optional, albeit desirable step.

LIST OF REFERENCE SIGNS

[0125] **1** Projection exposure apparatus
[0126] **2** Illumination system
[0127] **3** Radiation source
[0128] **4** Illumination optical unit
[0129] **5** Object field
[0130] **6** Object plane
[0131] **7** Reticle
[0132] **8** Reticle holder
[0133] **9** Reticle displacement drive
[0134] **10** Projection optical unit
[0135] **11** Image field
[0136] **12** Image plane
[0137] **13** Wafer
[0138] **14** Wafer holder
[0139] **15** Wafer displacement drive
[0140] **16** EUV radiation
[0141] **17** Collector
[0142] **18** Intermediate focal plane
[0143] **19** Deflection mirror
[0144] **20** Facet mirror
[0145] **21** Facets
[0146] **22** Facet mirror
[0147] **23** Facets
[0148] **30** Method step of "characterizing the actuator"
[0149] **31** Method step of "parameterizing the actuator model"
[0150] **32** Method step of "implementing the model"
[0151] **33** "Referencing" method step
[0152] **34** "Driving" method step
[0153] **101** Projection exposure apparatus
[0154] **102** Illumination system
[0155] **107** Reticle
[0156] **108** Reticle holder
[0157] **110** Projection optical unit
[0158] **113** Wafer
[0159] **114** Wafer holder
[0160] **116** DUV radiation
[0161] **117** Optical element
[0162] **118** Mounts
[0163] **119** Lens housing
[0164] M**1**-M**6** Mirrors

What is claimed is:

1. A method of driving an actuator for a component of a projection exposure apparatus for semiconductor lithography, the method comprising:
characterizing the actuator;
parameterizing an actuator model;
implementing the actuator model in a control structure; and
driving the actuator using the actuator model,
wherein characterizing the actuator is performed within the projection exposure apparatus.

2. The method of claim **1**, further comprising performing a reference step at certain times.

3. The method of claim **2**, wherein the reference step comprises a resetting of the model parameters.

4. The method of claim **2**, wherein the reference step comprises homing in on a defined actuator position.

5. The method of claim **2**, comprising performing the reference step between exposing a first wafer and a second wafer.

6. The method of claim **1**, wherein characterizing the actuator comprises detecting at least one parameter selected from the group consisting of a change in length, a frequency response, hysteresis, and a drift.

7. The method of claim **1**, further comprising generating at least one separate model for at least one of the actuator parameters, which is subsequently superposed on at least one further model, when parameterizing the actuator model.

8. The method of claim **1**, wherein the actuator comprises a member selected from the group consisting of an electrostrictive actuator, a piezoelectric actuator, and a magnetostrictive actuator.

9. The method of claim **1**, positioning the component using the actuator.

10. The method of claim **1**, deforming the component using the actuator.

11. The method of claim **1**, wherein the component comprises an optical element.

12. The method of claim **1**, wherein the component comprises a mirror.

13. The method of claim **1**, further comprising exposing a wafer in the projection exposure apparatus, wherein characterizing the actuator is based on image aberrations present during the wafer exposure.

14. The method of claim **13**, performing a reference step at certain times.

15. The method of claim **1**, further comprising exposing a wafer in the projection exposure apparatus, wherein characterizing a measurement of the exposed wafer.

16. The method of claim **15**, further comprising performing a reference step at certain times.

17. The method of claim **1**, further comprising performing a reference step at certain times wherein characterizing the actuator comprises detecting at least one parameter selected from the group consisting of a change in length, a frequency response, hysteresis, and a drift.

18. The method of claim **1**, further comprising:
performing a reference step at certain times; and
generating at least one separate model for at least one of the actuator parameters, which is subsequently superposed on at least one further model, when parameterizing the actuator model.

19. The method of claim **1**, further comprising performing a reference step at certain times, wherein the actuator comprises a member selected from the group consisting of an electrostrictive actuator, a piezoelectric actuator, and a magnetostrictive actuator.

20. The method of claim **1**, further comprising:
performing a reference step at certain times positioning the component using the actuator; and
positioning the component and/or deforming the component using the actuator.

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